

Prompt to Use with Claude Code / Cursor / Any AI Coding Assistant

Copy everything below into your AI coding tool along with the proposal document (upload it or paste its content first).

PROMPT START

I'm building a hackathon project called **PS8 - AI for Industrial Knowledge Intelligence** (a RAG-based chat assistant for industrial equipment documentation). I've attached the full proposal document with features, tech stack, and architecture.

Your task right now is ONLY setup — do not write feature logic yet. Do the following:

1. Create the folder/file structure

Set up a clean, production-style project skeleton based on the tech stack in the document (FastAPI backend, Chroma vector DB, PostgreSQL, React or Streamlit frontend — pick React unless I say otherwise). Structure should separate:

- `backend/` (FastAPI app, routes, services, models, db, ingestion pipeline, RAG logic)
- `frontend/` (React app, components, pages)
- `data/` (sample docs, uploads folder)
- `docs/` (project docs)
- Root-level config files (`.env.example`, `requirements.txt` or `pyproject.toml`, `package.json`, `.gitignore`, `README.md`)

Only create empty files / minimal boilerplate (folder placeholders, basic FastAPI "hello world" route, basic React app shell) — no real feature code yet. Confirm the structure with a tree view when done.

2. Create `PROGRESS_LOG.md`

A running log file at the project root. Every time you (the AI) complete any task — big or small — append an entry to this file in this format:

```
## [YYYY-MM-DD HH:MM] <Short task title>
- What was done: <1-3 lines>
- Files changed/created: <list>
- Status: Done / Partial / Blocked
- Notes: <anything I should know, decisions made, assumptions>
---
```

Do this automatically after every task without me having to ask. Never overwrite old entries — always append.

3. Create `FEATURE_CHECKLIST.md`

A checklist file at the project root, generated from the proposal document's feature list, organized like this:

markdown

PS8 Feature Checklist

MVP / Core Features

- ☐ Document upload (PDF, Word, Excel, scanned)
- ☐ OCR + text extraction pipeline
- ☐ Chunking + embedding + vector storage
- ☐ Hybrid search (vector + BM25)
- ☐ AI chat assistant with citations
- ☐ Equipment-aware retrieval

Differentiator Features

- ☐ Root cause analysis engine
- ☐ Temporal intelligence / failure timeline
- ☐ Predictive failure alerts
- ☐ Equipment relationship mapping
- ☐ Multi-document correlation
- ☐ Table/schematic extraction
- ☐ Cost impact analysis
- ☐ Compliance tracking

Dashboard

- ☐ Equipment health overview
- ☐ Alerts panel
- ☐ Failure timeline visualization
- ☐ Compliance status view

Polish / Stretch

- ☐ Error handling for bad uploads
- ☐ Demo data seeded
- ☐ Deployment-ready build

Whenever a feature is completed and tested, mark it `[x]` in this file **in the same turn** you finish the work — don't wait for me to ask. If a feature is partially done, add a `(in progress)` note next to it instead of checking it off.

4. Working rules going forward

For every task I give you from now on:

1. Do the task.
2. Update `FEATURE_CHECKLIST.md` if it relates to a listed feature.
3. Append an entry to `PROGRESS_LOG.md` describing what you did.
4. Then give me a short summary in chat (not the full log — just what changed).

Keep `PROGRESS_LOG.md` and `FEATURE_CHECKLIST.md` as the single source of truth for project status — I should be able to read just those two files at any point and know exactly where the project stands.

Confirm you understand this workflow, then proceed with the folder structure setup.

PROMPT END